# Classification of Target vs. Non-Target Fixations in Remote Sensing Imagery Using EEG and Eye-Tracking

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**Abstract**

Decoding target versus non-target fixations from neural signals remains challenging in visual search tasks. This study builds on the EEGET-RSOD dataset to classify individual eye fixations during expert search of remote sensing imagery using simultaneous EEG and eye-tracking data. Following rigorous quality control and preprocessing, we compared five modeling approaches, including deep learning architectures spanning both task-specific models and pretrained foundation backbones (EEGNet and CBraMod), Common Spatial Patterns, and feature-based Random Forest classifiers applied to both EEG and ET features, under within- and cross-subject validation schemes. Results show that a feature-based EEG classifier achieved the highest balanced accuracy on the pooled dataset (68.7%), while deep learning models struggled to generalize. Eye-tracking features performed comparably and excelled in cross-subject evaluation (66.7%), outperforming all EEG-based methods. Further analyses revealed that classification performance was modulated by visual search load, changing with target density, and that discriminative information was broadly distributed across cortical regions, with prominent centro-parietal contributions. Overall, these findings highlight both the challenges and opportunities in decoding fixation-related neural activity and suggest that multimodal approaches and methodological refinements are essential for reliable fixation-level intent decoding.

## 1 Introduction

Understanding how visual information can be decoded from neural signals is an important goal in neuroscience and applied machine learning, as it enables systems to infer task relevance and user intent without explicit responses. In visual search, this challenge arises at the level of eye fixations: observers fixate many objects, but only some are recognized as targets.

Decoding this distinction from gaze behavior alone is inherently limited. Eye tracking (ET) accurately captures where and when a person fixates and often provides strong cues about task relevance, but it does not directly indicate whether a fixated object has been cognitively recognized as a target. Electroencephalography (EEG) can provide complementary information related to internal processing such as target recognition, which is not explicitly encoded in gaze behavior. However, interpreting EEG in dynamic viewing conditions requires precise alignment with behavior to associate neural activity with specific visual events.

Combining EEG with eye tracking enables this alignment by linking neural activity to individual fixations. This fixation-aligned approach provides a principled framework for classifying fixations as target or non-target. Due to its high temporal resolution, EEG captures rapid neural responses around fixations that are relevant for this distinction. Prior work has shown that such multimodal approaches outperform EEG- or ET-only methods, particularly in complex and unconstrained environments [7].

In this context, He et al. [4] introduced the EEGET-RSOD dataset (Simultaneous EEG and Eye-Tracking Dataset for Remote Sensing Object Detection), which contains simultaneous EEG and eye-tracking recordings collected during free-viewing visual search in remote sensing images. The dataset includes recordings from 38 expert participants who were instructed to search for and fixate all targets within each image. Using fixation-locked EEG segments, the authors demonstrated that individual fixations can be classified as target or non-target based on neural activity, and that combining EEG and ET consistently improves fixation-level classification performance in complex scenes.

This study builds directly on the work of He et al. [4] by replicating and extending their fixation-level classification analysis using the same dataset. In addition to reproducing their results, we evaluate a broader set of EEG-only, ET-only, and multimodal models to assess how different modeling choices affect performance. We further analyze the influence of stimulus characteristics and examine whether neural signals from different brain regions contribute differently to fixation discrimination. Compared to prior work, we apply stricter data quality and preprocessing criteria to ensure robust evaluation.

# 2 Methods

## 2.1 EEGET-RSOD Dataset

The EEGET-RSOD dataset provides synchronized EEG and eye-tracking data from 38 remote sensing experts [4]. The dataset is provided as raw and preprocessed EEG signals, eye-tracking data with gaze points and labeled events (fixations, saccades, and blinks), the original stimuli, and bounding boxes for airplane locations.

Participants were selected based on a preliminary interpretation test, requiring over 90% accuracy in identifying objects within five seconds. The experiment presented 1,000 remote sensing images (800×800 pixels) from the DIOR dataset, including 900 target images containing one to six airplanes and 100 non-target control images. Example stimuli can be seen in Figure 1. Each image was displayed for 3,000 ms to better emulate natural visual search.

EEG was recorded using a 32-channel NE Enobio system at 500 Hz following the 10-20 system, while gaze data were captured with an SMI RED250 eye tracker at 250 Hz with 0.03° spatial resolution. Temporal alignment between modalities, a major challenge in EEG and eye-tracking studies, was facilitated using keyboard-input timestamps at the start and end of each block. Technical validation confirmed dataset robustness, with participants achieving an average object detection rate of 90.11%. Event-related potential analyses revealed consistent P300 (Pz) and N1 (Cz) components, reflecting attention and visual processing during the task [4]. The dataset is publicly available for academic use at `https://figshare.com/articles/dataset/EEGET-RSOD/26943565`.

## 2.2 Data Preprocessing

**Fixation Annotation and Labeling** Fixations were labeled as on-target or off-target following the procedure of [4]. Binocular eye-tracking data were combined into a single gaze representation, and fixation events were taken from the provided labels. Fixation locations were defined as the mean gaze position over each event. Object locations were obtained from the DIOR dataset using airplane bounding boxes from the corresponding XML annotations. To account for the limited effective visual field, each fixation was modeled as a circular region with a 50-pixel radius around its center. A fixation was labeled as "on airplane" if this region overlapped with any airplane bounding box; otherwise, it was labeled as "off airplane."

Although this procedure closely follows [4], the resulting object detection rates were generally lower than those reported in the original paper. This discrepancy likely arises from unspecified

details in the original eye-tracking processing, including how binocular gaze signals were combined and how gaze samples were aggregated into fixations. In our implementation, left and right eye gaze positions were averaged, which may differ from the original processing. Furthermore, while fixation detection criteria are described in the paper, the dataset may include fixation labels generated by the eye tracker using a different algorithm. Such differences can affect fixation locations and, consequently, object overlap and detection rates. Nevertheless, visual inspection of gaze traces and fixation timing indicated that, in most trials, the provided labels and event timings were plausible and consistent with recorded gaze behavior.

**EEG Preprocessing and Epoch Labeling** Preprocessed EEG data provided by the dataset were used for subsequent analysis. Preprocessing included channel location assignment, high-pass filtering at 0.1 Hz, low-pass filtering at 80 Hz, a 50 Hz notch filter, average referencing, and artifact removal using independent component analysis (ICA) with manual exclusion of eye-movement components.

To align EEG with eye-tracking data, a temporal offset parameter provided by the dataset was applied, accounting for differences in device timing and sampling rates (see [4]). This alignment enabled precise mapping of fixation events onto EEG recordings.

Using this alignment, time-locked EEG epochs were extracted around each fixation, and baseline normalization was applied using a 300 ms pre-fixation interval to reduce slow drifts. Epochs with inconsistent segment lengths or missing/invalid fixation labels were excluded to ensure data integrity. The resulting epochs, together with fixation labels, stimulus information, channel names, sampling rate, and time axis, were stored per subject as compressed files. These labeled EEG segments formed the basis for subsequent fixation-level classification analyses.

### 2.3 Data Quality

To ensure high reliability in our analysis, we applied a rigorous, manual quality control pipeline to the original dataset, focusing on eye-tracking (ET) validation, participant engagement, and EEG–ET synchronization. Initial inspection of gaze and stimulus data revealed inconsistencies in calibration for certain trials, characterized by shifted gaze points or fixations recorded outside the stimulus boundaries despite the participant's apparent engagement. Figure 1 illustrates examples of well-calibrated trials that were included in the final dataset, whereas Figure 2 shows trials with poorer calibration that were excluded.

To mitigate these issues, strict inclusion criteria were implemented. A trial was considered valid only if it was a non-target trial (containing no airplanes) or, for plane trials, if all targets were successfully fixated ($n_{targets} == n_{fixated}$) and the participant registered both target and non-target fixations ($n_{on} > 0$ and $n_{off} > 0$). This filtering eliminated trials affected by poor calibration and verified participant compliance; given the relative simplicity of the visual search task, requiring all targets to be fixated ensured that participants were attentive and actively searching.

From the pool of valid trials, a stratified selection strategy was employed to standardize dataset size and maximize data quality. For each subject, all valid non-target trials were retained to preserve dataset heterogeneity. Additional target trials were then selected by prioritizing those with the highest engagement contrast, ranking them by the absolute difference between on-target and off-target fixations ($|n_{on} - n_{off}|$). The top-ranked trials were retained until a threshold of 250 trials per subject was reached. This reduction homogenized data volume across subjects, mitigating statistical bias while maximizing the signal-to-noise ratio by prioritizing trials with distinct fixation patterns. Subjects failing to meet the minimum threshold of 250 valid trials were excluded.

Furthermore, the temporal alignment between EEG and ET data was validated through manual inspection by cross-referencing blinks detected in the ET stream with corresponding

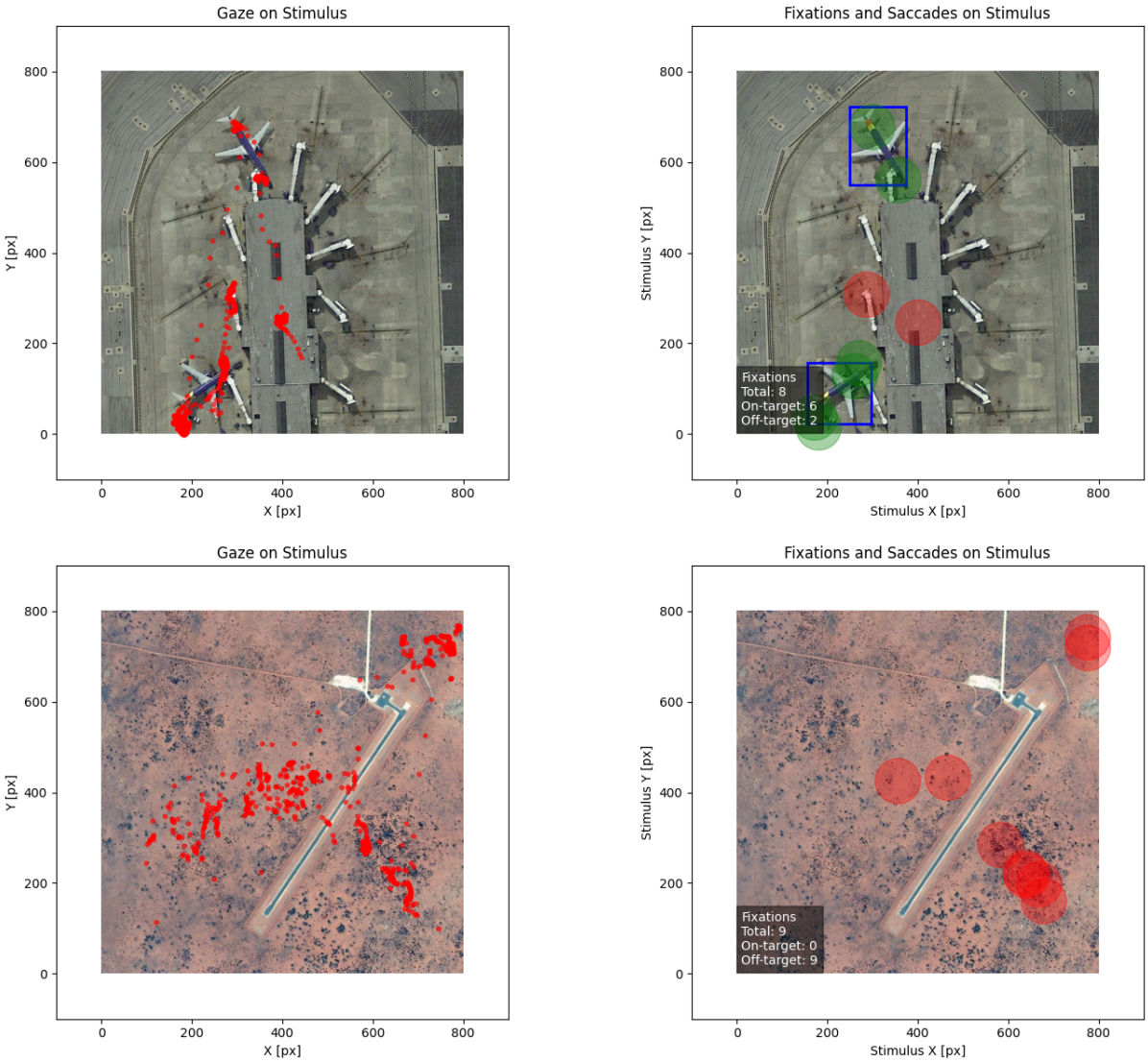



Figure 1: Eye-tracking examples for subject P01 demonstrating good calibration and fixation detection. Left column: raw gaze points; right column: detected fixations (green: target, red: non-target). Blue bounding boxes indicate airplane locations. Top row: stimulus with airplanes; bottom row: stimulus without airplanes.

artifacts in the raw EEG signal (Figure 3). This process identified subjects with severe signal degradation, including unaligned epochs or electrodes showing negligible voltage activity. To prevent the introduction of invalid epochs or incorrect labels, only subjects demonstrating clear, artifact-verified alignment were retained. The final quality-controlled dataset comprised 13 participants, resulting in a total of 33,012 fixations used for classification and ERP visualization. A detailed breakdown of fixation distribution per subject is provided in Table 1.

## 2.4 Models

Five models were used to classify whether a fixation made by a subject was on-target (an airplane) or off-target (not on an airplane): EEGNet, CBraMod, CSP, a feature-based classifier using hand-engineered EEG features, and a classifier based solely on eye-tracking (ET) features. Each model was trained and evaluated using procedures appropriate to its architecture, while maintaining consistent principles to allow a fair comparison of performance.

For all models, performance was evaluated using two complementary validation schemes.

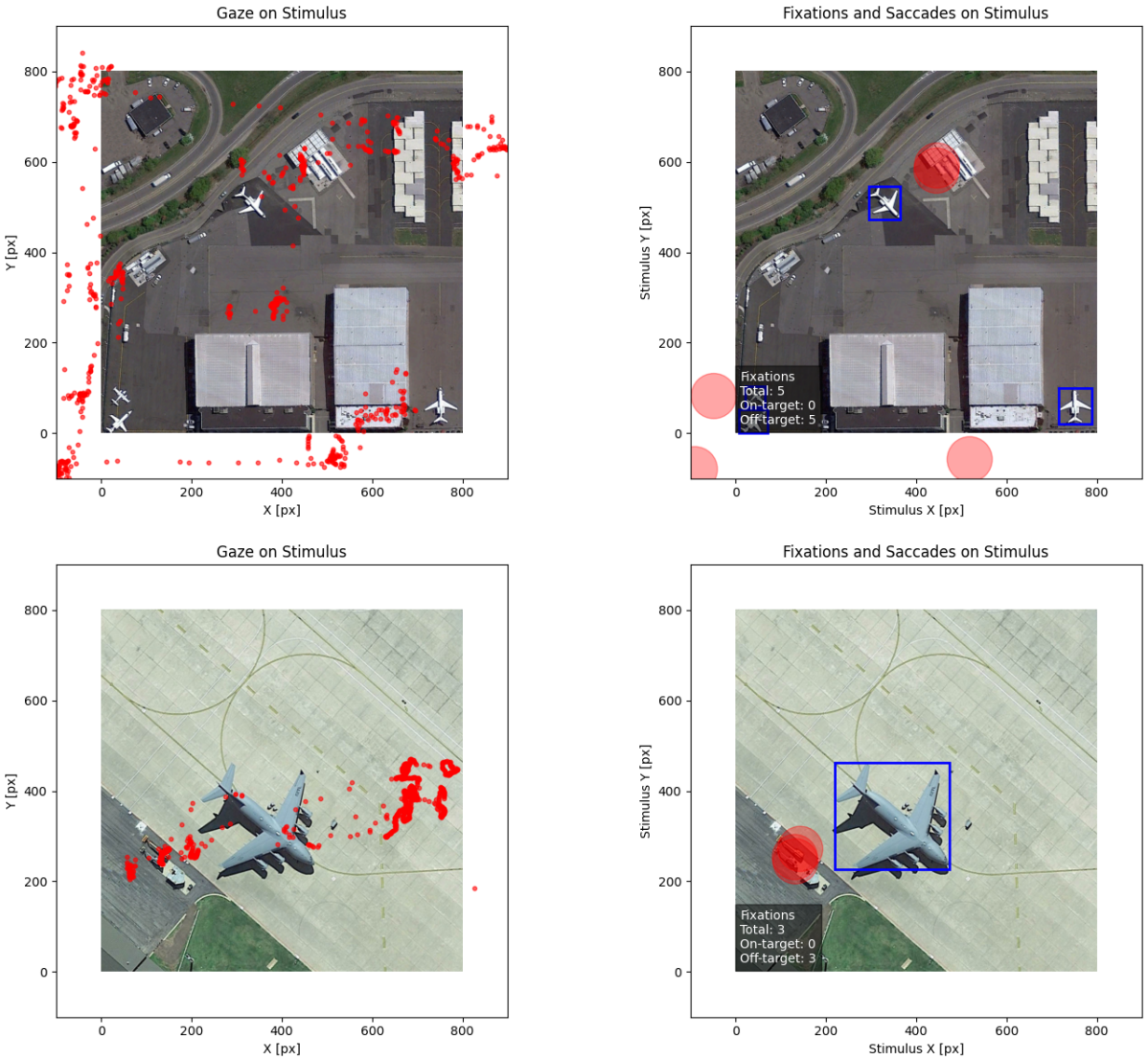



Figure 2: Eye-tracking examples for subject P01 demonstrating poorer calibration and fixation detection. Left column: raw gaze points; right column: detected fixations (green: target, red: non-target). Blue bounding boxes indicate airplane locations. Top row: stimulus with many distractors; bottom row: stimulus with a single airplane.

Pooled validation involved randomly splitting all trials from all participants into training, validation, and test sets (60/20/20), ignoring subject identity, while cross-subject validation held out three participants for testing and used the remaining ten for training. The validation sets were used for hyperparameter tuning to prevent data leakage and ensure that test performance reflects unbiased generalization. Balanced accuracy was selected as the primary performance metric to account for class imbalance between target and non-target fixations. By averaging the recall for each class, this metric assigns equal weight to both classes, preventing results from being dominated by the more frequent non-target fixations. EEG-based analyses were restricted to 19 channels (Fp1, Fp2, F7, F3, Fz, F4, F8, T7, C3, Cz, C4, T8, P7, P3, Pz, P4, P8, O1, O2) to ensure comparability across models and compatibility with CBraMod's pre-trained input configuration. These channels cover the major cortical regions involved in visual attention and fixation processing [8]. Model-specific preprocessing, including filtering, epoch segmentation, and scaling, was applied according to each architecture's requirements. Class balancing strategies were employed where necessary to mitigate residual differences in the number of target

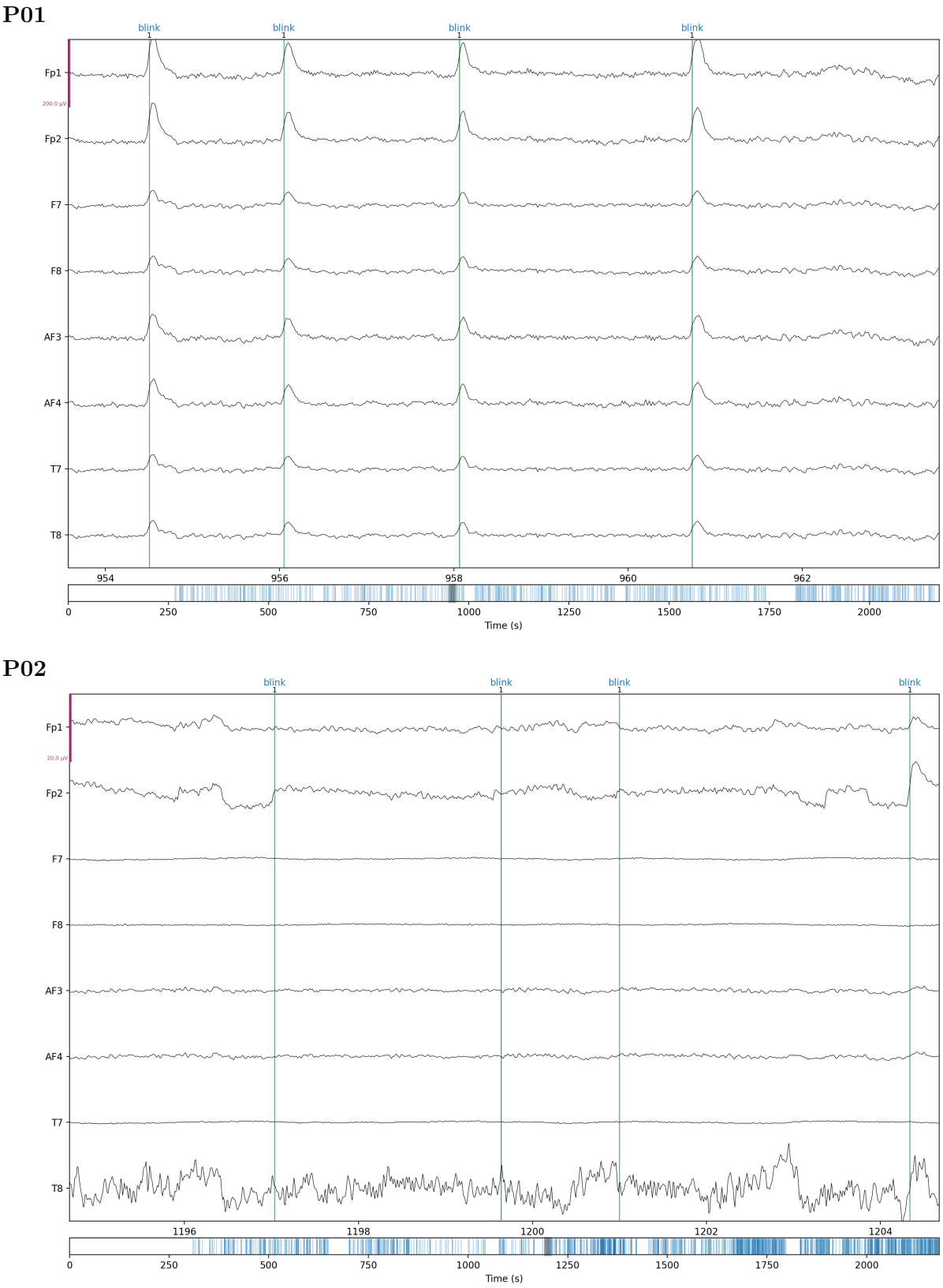



Figure 3: Time alignment of EEG and eye-tracking data for participants P01 and P02. Raw EEG segments are shown for handpicked intervals. Vertical lines indicate blinks detected by the eye tracker. P01 (included in the final dataset) shows EEG amplitudes in the range of 200 µV, whereas P02 waves (excluded) are shown in a lower range ( 20 µV).

| Subject | Total Fixations | ON Target (%) | OFF Target (%) |
|---|---|---|---|
| P01 | 2452 | 1257 (51.3%) | 1195 (48.7%) |
| P07 | 2509 | 1391 (55.4%) | 1118 (44.6%) |
| P09 | 2218 | 1137 (51.3%) | 1081 (48.7%) |
| P10 | 2646 | 1490 (56.3%) | 1156 (43.7%) |
| P16 | 2829 | 1785 (63.1%) | 1044 (36.9%) |
| P17 | 3051 | 1803 (59.1%) | 1248 (40.9%) |
| P20 | 2487 | 1306 (52.5%) | 1181 (47.5%) |
| P21 | 3085 | 1716 (55.6%) | 1369 (44.4%) |
| P27 | 2675 | 1526 (57.0%) | 1149 (43.0%) |
| P28 | 2121 | 1062 (50.1%) | 1059 (49.9%) |
| P30 | 2193 | 1275 (58.1%) | 918 (41.9%) |
| P34 | 2673 | 1439 (53.8%) | 1234 (46.2%) |
| P38 | 2073 | 1105 (53.3%) | 968 (46.7%) |
| **Total** | **33,012** | **18,292 (55.4%)** | **14,720 (44.6%)** |
| **Average** | **2539** | **1407** | **1132** |

Table 1: Distribution of Fixations for Selected Participants.

versus non-target fixations.

The following subsections describe each model in detail, including its role, architecture, preprocessing pipeline, and hyperparameter configuration.

#### 2.4.1 EEGNet

EEGNet extracts spatiotemporal features from EEG to classify target versus non-target fixations. Following the experimental protocol of the original dataset study [4], EEGNet [5] is a compact convolutional neural network designed for EEG-based brain–computer interface applications. Its architecture combines temporal convolutions with depthwise and separable spatial convolutions, efficiently capturing frequency-specific temporal features and inter-channel spatial patterns while maintaining a low parameter count, making it suitable for datasets with limited training samples.

EEG inputs were subject-wise Z-score normalized to reduce inter-subject amplitude variability and stabilize optimization. The network employed ELU activations throughout and was trained using the Adam optimizer with a batch size of 32, learning rate of 0.0005, and early stopping based on validation loss to prevent overfitting.

The optimized model configuration included $F_1 = 16$ temporal filters and depth multiplier $D = 4$, controlling temporal feature richness and the number of spatial filters per temporal filter, respectively. The second convolutional layer used $F_2 = 16$ filters, and a dropout rate of 0.3 was applied. All remaining components followed the original EEGNet design, providing a consistent and interpretable baseline for fixation classification.

### 2.4.2 CBraMod

CBraMod leverages pre-trained EEG representations to classify target versus non-target fixations [8]. Unlike conventional EEG decoding models, which are trained from scratch and often tied to a specific dataset, CBraMod learns transferable representations through a criss-cross transformer backbone that models spatial (inter-channel) and temporal dependencies using parallel attention mechanisms. This architecture is well suited to heterogeneous EEG signals.

The model was initialized with the official pre-trained weights provided by the authors, obtained from large-scale pre-training on the Temple University Hospital EEG Corpus (TUEG). Pre-training used a patch-based masked EEG reconstruction objective, enabling the model to learn generic, task-agnostic representations across multiple subjects. Prior studies have demonstrated strong transfer performance on diverse EEG classification tasks, motivating the use of these weights for fixation-level classification.

For binary classification, the original reconstruction head was replaced with a task-specific MLP consisting of a flattening layer, two fully connected layers with ELU activations and dropout, and a final sigmoid-activated output neuron. During fine-tuning, all pre-trained weights were updated, as freezing the backbone degrades performance.

EEG signals were downsampled to 200 Hz, segmented into 1-second windows, and normalized by 100 µV to match the pre-training regime. Data were reshaped into the required spatial–temporal patch format ($C \times n \times t$). Fine-tuning used a batch size of 64 with the AdamW optimizer, learning rate of $5x10^{-5}$, weight decay of 0.05, dropout of 0.1, hidden dimension of 512, and label smoothing set to 0.

### 2.4.3 Common Spatial Patterns

Common Spatial Pattern (CSP) was applied to EEG epochs to extract spatial features that differentiate target from non-target fixations [7]. CSP computes class-specific covariance matrices to derive spatial filters that maximize variance for one class while minimizing it for the other, isolating scalp distributions most informative for fixation-level discrimination.

EEG epochs were first Z-score normalized per subject to reduce inter-subject amplitude variability and stabilize covariance estimation. The resulting CSP features were further standardized using a separate scaler to ensure consistent ranges for classifier training. CSP was configured with 15 spatial components and Ledoit-Wolf regularization of 0.05, balancing variance capture with regularization while preserving discriminative information.

Classification was performed using a Random Forest (RF), which captures non-linear patterns in the CSP feature space and outperformed linear alternatives such as Linear Discriminant Analysis (LDA). This two-step approach, CSP for feature extraction and RF for classification, should enable robust discrimination between target and non-target fixations while keeping feature extraction and classification distinct.

### 2.4.4 Feature-based EEG classification

Hand-engineered EEG features were extracted from each fixation interval to implement a feature-based classification pipeline. The feature set was largely based on the study of Sharma et al. [7], reflecting commonly used signal-processing and statistical metrics in EEG analysis. Features included frequency-domain measures (e.g., power spectral density across standard EEG bands), fractal and complexity indices (e.g., Petrosian and Higuchi fractal dimensions, Hjorth mobility and complexity, detrended fluctuation analysis), and statistical descriptors (e.g., mean, standard deviation, skewness, kurtosis, minimum, maximum), along with additional time-domain characteristics.

All features were standardized per subject to account for inter-subject amplitude differences and stabilize classifier training. Multiple classifiers were evaluated, including SVM and RF, with

RF consistently outperforming SVM. The final RF configuration used 200 estimators, maximum depth of 30, minimum samples per split of 2, and minimum samples per leaf of 1.

#### 2.4.5 Eye-tracking feature-based classification

Classification based solely on eye-tracking features was used to distinguish target and non-target fixations without relying on EEG. Features included fixation duration, pupil size (mean, standard deviation, range), gaze position and dispersion, and saccade-related metrics preceding and following each fixation. Previous studies have shown that eye-tracking features alone can achieve high accuracy in fixation-based tasks [7, 1].

Features were standardized per subject to reduce inter-individual variability. Both SVM and RF classifiers were tested, with RF demonstrating superior performance. Hyperparameter tuning identified the optimal RF configuration as 200 estimators, maximum depth of 30, minimum samples per split of 5, and minimum samples per leaf of 2.

## 3 Results

### 3.1 Event-Related Potentials (ERPs)

**Stimulus-locked ERPs** To assess the quality of the EEG recordings and characterize the neural response to stimulus onset, stimulus-locked ERPs were analyzed, focusing on the parietal midline electrode Pz. This site captures the P300 component, a well-established marker of attentional engagement and context updating, while also able to indicate other ERPs like the N100, reflecting early sensory processing and attentional orienting.

The grand-average waveform (Figure 4) shows a clear biphasic response. A small negative deflection around 80–120 ms corresponds to the N100, reflecting early sensory registration of the visual stimulus and the initial allocation of attention. This is followed by a prominent positive deflection peaking between 200 and 400 ms, corresponding to the P300. The P300 amplitude indexes the degree to which cognitive resources are allocated to stimulus evaluation and working memory updating, with larger amplitudes reflecting greater attentional engagement.

Although the waveform differs from that reported by He et al. [4], likely due to preprocessing differences or sample size, the presence of a clear P300 and identifiable N100 confirms the temporal validity of the recordings. These findings indicate that participants were attentive to the stimuli and engaged the neural mechanisms typically associated with early sensory processing and higher-level attentional evaluation.

**Fixation-locked ERPs** To examine neural responses associated with target versus non-target fixations, ERPs were extracted time-locked to fixation onset. Channels Cz and Pz were selected to illustrate how different scalp locations exhibit condition-dependent differences in the EEG signal, providing insight into the spatial distribution of fixation-related activity. The grand-average waveforms (Figure 5) show that for channel Cz there is a clear divergence between target and non-target fixations, with a larger positive deflection for targets in the 200–400 ms window, consistent with a P300-like response. This target-related positivity is less pronounced at channel Pz, where differences between fixation types are minor.

These observations suggest that certain channels, such as Cz, may carry more information relevant to distinguishing target from non-target fixations, whereas others, like Pz, show only subtle differences. This highlights how channel selection can influence the detectability of condition-specific EEG patterns and may inform the choice of features for subsequent classification analyses.

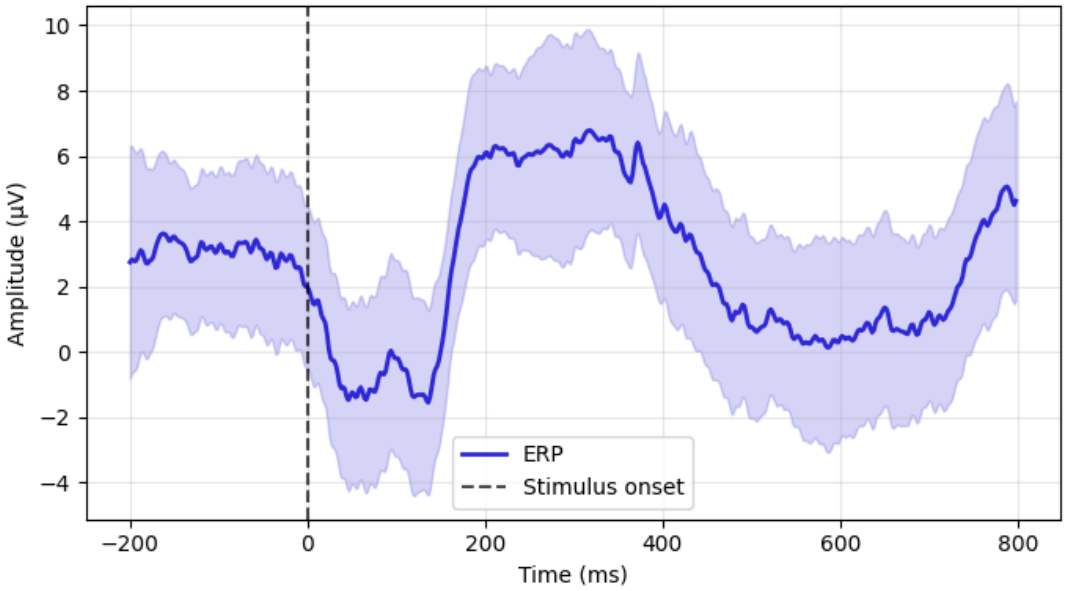
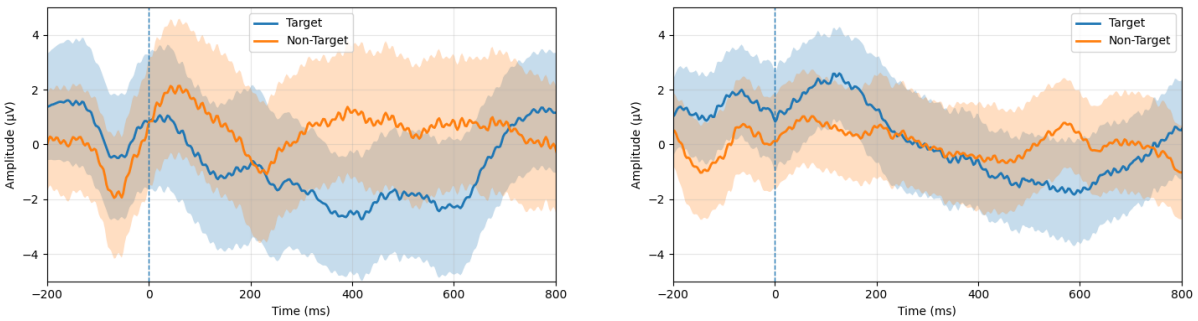



Figure 4: Grand-average ERP waveform at Pz, time-locked to stimulus onset (0 ms). The solid line shows the mean amplitude ($\mu V$) across participants; the shaded area indicates the 95% confidence interval.

### 3.2 Classification Performance

Fixation classification performance varied across models and validation schemes. Under pooled validation (Table 2), the feature-based EEG classifier achieved the highest balanced accuracy at 68.7%, showing that hand-engineered EEG features capture neural signatures relevant for distinguishing target from non-target fixations between participants. The eye-tracking (ET) classifier performed similarly (67.2%), demonstrating that gaze and pupil dynamics alone provide reliable information for fixation relevance. CSP, CBraMod, and EEGNet performed substantially worse. This suggests that both traditional spatial filtering (CSP) and deep learning approaches struggle to extract discriminative patterns from raw EEG without subject-specific adaptation. CSP's moderate performance indicates some utility of variance-based spatial features, while EEGNet and CBraMod might be limited by small dataset size, high inter-trial variability, and subtle fixation-related EEG components.




(a) Channel Cz

(b) Channel Pz

Figure 5: Grand-average ERP waveforms for target (blue) and non-target (orange) fixations at electrodes Cz (a) and Pz (b). Signals are time-locked to fixation onset (0 ms). Shaded regions indicate 95% confidence intervals. Channels were chosen to illustrate differences in EEG responses between target and non-target fixations across scalp locations.

Cross-subject validation highlighted these differences further (Table 3). The ET classifier maintained strong performance (66.7%) on unseen subjects, indicating robust generalization of eye-tracking features. In contrast, all EEG-based models dropped to near-chance levels. These results underscore the challenges of generalizing EEG-based fixation classifiers across individuals, where variability in amplitude, electrode placement, and cognitive strategy reduces reliability. Even CBraMod, despite large-scale pre-training, did not improve performance, suggesting that fine-tuning on small fixation datasets is insufficient to overcome subject-specific differences.

Overall, EEG features are informative if seen before but do not generalize well across subjects, while eye-tracking features are consistently robust. The limited performance of deep learning models emphasizes the difficulty of applying high-capacity architectures, high-variability EEG datasets, highlighting the potential need for stronger regularization, or multimodal integration for robust classification.

| Model | Balanced Accuracy (%) |
|---|---|
| Feature-based EEG (RF) | 68.7 |
| Eye-tracking (RF) | 67.2 |
| CSP (RF) | 56.3 |
| CBraMod | 55.3 |
| EEGNet | 53.8 |

Table 2: Balanced accuracy for each model using pooled validation. Models were trained and tested on random splits of the aggregated dataset containing fixations from all subjects.

| Model | Balanced Accuracy (%) |
|---|---|
| Feature-based EEG (RF) | 50.6 |
| Eye-tracking (RF) | 66.7 |
| CSP (RF) | 49.5 |
| CBraMod | 49.9 |
| EEGNet | 50.2 |

Table 3: Balanced accuracy for each model using cross-subject validation (leave-three-subjects-out). Models were trained on 10 subjects and evaluated on three held-out subjects.

### 3.3 Common Spatial Pattern Analysis

A CSP model with four components was trained to identify spatial patterns that distinguish target from non-target fixations. Topographical maps of these components (Figure 6) illustrate the relative contribution of each electrode, with red and blue indicating strong weights of opposite polarity and lighter colors indicating weaker contributions. Polarity reflects how the model contrasts signals across regions to maximize variance for one class while minimizing it for the other.

Components 1 and 2, which maximize variance for non-target fixations, show stronger weights over posterior-occipital and lateral-parietal electrodes. Component 3 displays weak or inconsistent patterns. Component 4, which maximizes variance for target fixations, highlights the centro-parietal midline near Cz and Pz, consistent with regions associated with the P300 response. This pattern aligns with the fixation-locked ERPs shown in Figure 5, indicating that CSP captured variance related to target-specific neural activity.

It is important to note that CSP identifies patterns of variance rather than exact localized activity. The moderate classification accuracy (54%) suggests that fixation-related neural sig-

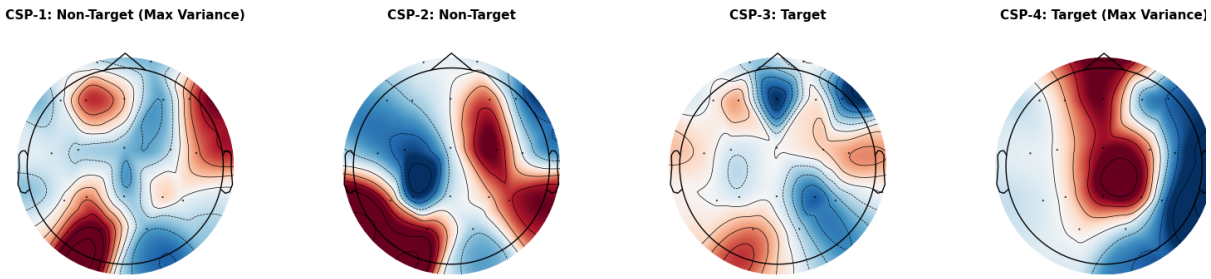



Figure 6: Topographical maps of the four CSP components from the trained model. Each component represents a spatial filter that maximizes variance for one class while minimizing it for the other. Red and blue indicate high-magnitude weights of opposite polarity at each electrode, with lighter colors showing weaker contributions, highlighting how different scalp regions influence each component.

nals are subtle and variable. Nevertheless, the prominence of P300-related electrodes in the target-maximizing component highlights neural regions contributing most to distinguishing target fixations.

### 3.4 Effect of Visual Search Load on Classification Performance

To examine whether visual search load influenced model performance, classification accuracy was analyzed as a function of target density, defined by the number of airplanes present in each stimulus. Stimuli with no targets may require more exhaustive searching, as participants need to confirm the absence of relevant objects, while scenes with multiple targets require sustained search and attentional engagement to locate each target. In contrast, images with only one or two targets may be easier to process, potentially eliciting weaker or more variable neural responses.

Models were trained on the full dataset and evaluated post hoc using balanced accuracy computed separately for each target-density condition. As shown in Table 4, performance varied across conditions, ranging from 68.4% to 75.5%. These differences, while not conclusive, are supported by the relatively large number of trials in most conditions and may reflect genuine effects of target density on fixation-related classification.

Accuracy was highest for stimuli without targets, likely reflecting the combined effect of participants' careful search and the reliable detection of non-target trials. For target-present stimuli, performance was lowest for scenes with one or two airplanes and tended to increase for higher target densities (three to six airplanes). This pattern could indicate that both very low and very high target densities elicit stronger or more consistent neural responses, whereas minimally engaging stimuli with only a few targets produce weaker signals. Overall, these results suggest that visual search demands modulate neural responses relevant for fixation classification, though the effect is subtle and should be interpreted with caution.

### 3.5 Regional Performance and Functional Mapping

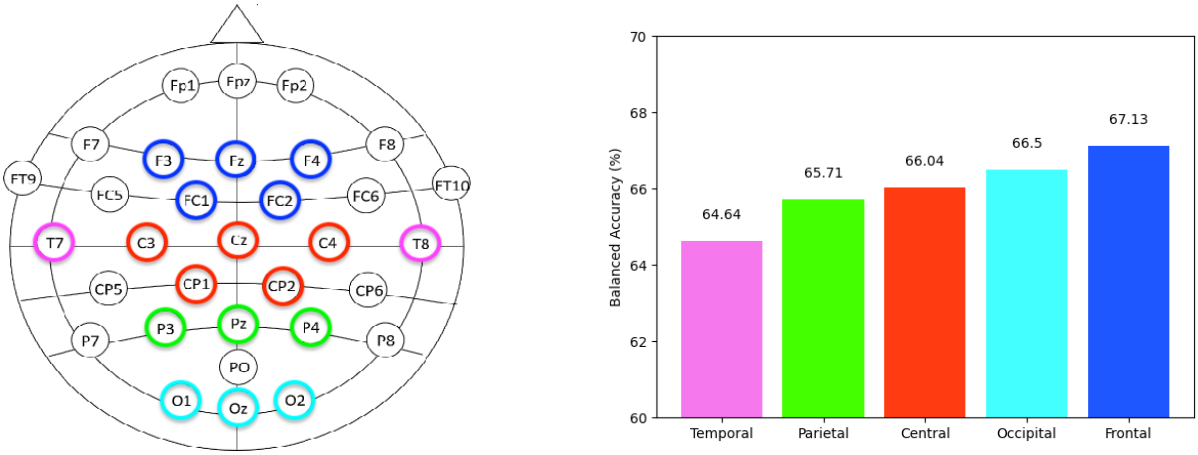
To explore whether specific cortical regions contribute differentially to fixation classification, EEG channels were grouped according to their correspondence with major functional networks relevant to visual target detection and goal-directed behavior. The clusters were defined as follows (Fig. 7a): Occipital (Oz, O1, O2) for early visual processing; Parietal (P3, P4, Pz) overlying the dorsal attention network for spatial selection; Temporal (T7, T8) for ventral-stream object recognition; Frontal and fronto-central (F3, F4, FC1, FC2, Fz) for executive control; and Central/centro-parietal (C3, C4, CP1, CP2, Cz) for sensorimotor and response-related activity [6, 3].

| N-Airplanes | Balanced Accuracy | Trials ($n$) |
|---|---|---|
| 0 | 0.755 | 1337 |
| 1 | 0.684 | 1628 |
| 2 | 0.685 | 1851 |
| 3 | 0.713 | 732 |
| 4 | 0.699 | 592 |
| 5 | 0.743 | 300 |
| 6 | 0.747 | 146 |

Table 4: Balanced classification accuracy for stimuli grouped by the number of airplanes.

Regional classification performance (Figure 7) was relatively uniform, with balanced accuracy ranging from 64.6% to 67.1%. The Frontal cluster achieved the highest accuracy, consistent with the P300-related contributions identified in CSP component 4, while Occipital electrodes were second, reflecting the role of early visual processing. Temporal electrodes, represented by only two channels, showed the lowest accuracy, possibly reflecting both limited coverage and lower task-relevant signal. The modest difference across regions (∼2.5%) suggests that discriminative neural information is distributed rather than confined to a single area.

Electrode density may also influence these results: clusters with more channels, such as Frontal and Central regions, may benefit from higher signal-to-noise ratios and a richer feature space. Overall, all regions carry informative signals for fixation classification, with frontal, central, and occipital areas contributing most prominently.




(a) Regional electrode groupings.

(b) Balanced accuracy by region.

Figure 7: Regional EEG analysis. (a) Electrode clusters mapped to functional regions: Frontal (blue), Central (red), Temporal (pink), Parietal (green), and Occipital (cyan). (b) Balanced accuracy for each cluster, showing relatively uniform performance across regions, with frontal, central, and occipital areas contributing most strongly to fixation classification. Differences are exploratory and not indicative of precise localization.

## 4 Discussion

The results of this study highlight a clear performance disparity between feature-based and deep learning approaches for classifying target versus non-target fixations in remote sensing images. The feature-based Random Forest classifier consistently outperformed all other models, likely because hand-engineered EEG features compress the high-dimensional neural signal and reduce

sensitivity to noise. The per-subject scaling and the ensemble structure of the model further mitigate variance, making it robust even with a relatively small dataset. In contrast, deep learning models, including EEGNet and CBraMod, achieved high training accuracy but failed to generalize, even in the pooled condition. It is worth noting that despite the large volume of individual fixation events ($N = 33,012$), the effective sample size for learning generalizable neural patterns remains constrained by the participant count ($N = 13$). Deep learning architectures like EEGNet and CBraMod typically require higher subject diversity to overcome the idiosyncratic signal topographies that characterize individual EEG recordings.

The poor cross-subject performance is likely due to the high signal-to-noise ratio in fixation-locked EEG, limited training samples, and especially individual variability in skull geometry, cortical folding, and electrode impedance, all of which produce unique signal topographies across participants. Notably, the eye-tracking classifier proved highly effective and robust, performing comparably to the best EEG models within subjects and significantly better across subjects. This strong performance suggests that oculomotor dynamics provide a simpler and more reliable signal for target detection in this context, raising questions about the necessity of EEG for this particular task.

The nature of the visual search task in the EEGET-RSOD dataset introduces additional challenges. Unlike naturalistic scenes used in other studies, the satellite images are dense, top-down views containing partly huge airplane targets. We hypothesize that participants adopt a rapid scanning or glancing strategy rather than doing an extended search. Big targets can often be detected via peripheral vision, resulting in brief fixations that may not even lead to an fixation and not elicit a clear ERP before the next eye movement occurs, thereby blurring the distinction between target and non-target signals. This effect is further modulated by target density. Stimuli without targets were unexpectedly engaging, as participants had to actively verify the absence of airplanes, yielding a relatively consistent non-target neural signature. In contrast, images containing only one or two targets required little sustained search, resulting in reduced engagement, more variable neural responses, and lower classification accuracy. As target count increased, repeated interactions with multiple targets promoted more sustained search behavior, producing stronger and more consistent neural signals and, consequently, slightly higher accuracy. This trend underscores the importance of active visual search, rather than brief glancing, for reliable neural discrimination in these trials.

Regional analyses showed that classification performance was relatively uniform across the scalp, indicating that discriminative neural information is broadly distributed rather than confined to a single cortical area. However, Common Spatial Pattern (CSP) analysis provided neurophysiological insight, revealing that variance associated with target fixations was most prominent at centro-parietal sites. These locations correspond to the P300 component observed in fixation-locked ERPs, suggesting that the classifier leveraged meaningful task-related neural activity rather than spurious signals.

Despite these insights, several limitations constrain the current study. Free-viewing EEG inherently involves overlapping neural responses from consecutive fixations, which reduces the signal-to-noise ratio for any single interval. Future work should address this by applying overlap correction or deconvolution methods, such as those proposed by Dimigen and Ehinger [2], to isolate the pure neural response associated with each fixation. Moreover, combining EEG and eye-tracking into a hybrid fusion model could exploit the strengths of both modalities, pairing the temporal precision of cortical signals with the cross-subject robustness of oculomotor features. There is also substantial potential for further optimization of the models themselves. Feature importance analyses could clarify which aspects of the EEG signal drive Random Forest classification, while CBraMod could benefit from a more specialized downstream classifier better suited for binary discrimination under noisy conditions. Exploring different epoch intervals, increasing the dataset size, and fine-tuning model hyperparameters even further could improve performance.

In conclusion, this dataset presents a particularly challenging scenario due to the small, dense stimuli, brief fixations, and moderate EEG quality. The task encourages rapid scanning rather than prolonged visual search, which may attenuate neural signatures. Nevertheless, classification was possible, and the relationship between target-related activity and accuracy provides a meaningful neurophysiological validation of the models. While overall performance was lower than reported in prior studies using more traditional search tasks, this work establishes a solid foundation for future investigation and model refinement. It highlights both the challenges and opportunities inherent in decoding fixation-related neural activity in complex remote sensing imagery.

## References


[1] Anne-Marie Brouwer, Maarten A Hogervorst, Bob Oudejans, Anthony J Ries, and Jonathan Touryan. Eeg and eye tracking signatures of target encoding during structured visual search. *Frontiers in human neuroscience*, 11:264, 2017.

[2] Olaf Dimigen and Benedikt V Ehinger. Regression-based analysis of combined eeg and eye-tracking data: Theory and applications. *Journal of vision*, 21(1):3–3, 2021.

[3] E Guttmann-Flury, X Sheng, and X Zhu. Channel selection from source localization: A review of four eeg-based brain–computer interfaces paradigms. *Behavior Research Methods*, 55(4):1980–2003, 2023.

[4] Bing He, Hongqiang Zhang, Tong Qin, Bowen Shi, Qiao Wang, and Weihua Dong. A simultaneous eeg and eye-tracking dataset for remote sensing object detection. *Scientific Data*, 12(1):651, 2025.

[5] Vernon J Lawhern, Amelia J Solon, Nicholas R Waytowich, Stephen M Gordon, Chou P Hung, and Brent J Lance. Eegnet: a compact convolutional neural network for eeg-based brain–computer interfaces. *Journal of neural engineering*, 15(5):056013, 2018.

[6] John Polich. Neuropsychology of p300. *The Oxford handbook of event-related potential components*, 641:159–188, 2012.

[7] Mansi Sharma, Camilo Andrés Martínez Martínez, Benedikt Emanuel Wirth, Antonio Krüger, and Philipp Müller. Distinguishing target and non-target fixations with eeg and eye tracking in realistic visual scenes. In *Proceedings of the 26th International Conference on Multimodal Interaction*, pages 459–468, 2024.

[8] Jiquan Wang, Sha Zhao, Zhiling Luo, Yangxuan Zhou, Haiteng Jiang, Shijian Li, Tao Li, and Gang Pan. Cbramod: A criss-cross brain foundation model for eeg decoding. *arXiv preprint arXiv:2412.07236*, 2024.


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